



## 1. What is a Robot?

**Core Definition:** An electromechanical system characterized by three essential traits:

- **Reprogrammable** Flexible software control
- **Multifunctional** Versatile multi-task capability
- **Sensible** Real-time environment awareness

### Key Terminology:

- **Robot:** Physical mechanism performing human tasks autonomously or via teleoperation.
- **Robotics:** Science of mechanical design, kinematics, control & software.
- **Telerobotics:** Remote-controlled operation in extreme/hazardous environments.

## 2. Automation vs. Robots

### Fixed Automation

- **Single dedicated task**
- Maximum speed & efficiency
- Rigid; high retooling cost
- Ex: Bottling line, dishwasher

### Flexible Robots

- **Diverse multiple tasks**
- Reprogrammable motions
- High mechanical flexibility
- Ex: 6-DOF arms, AGVs, CNCs

## 3. Asimov's 3 Laws of Robotics

- 1 **Safety First:** May not injure a human or, through inaction, allow harm.
- 2 **Obedience:** Must obey human orders, **except** when conflicting with Law 1.
- 3 **Self-Defense:** Must protect own existence, unless conflicting with Laws 1 or 2.

## 4. Core Hardware Architecture

- Sensors** **Data Input:** Internal state (encoders, resolvers) & external world (cameras, LIDAR, tactile/touch).
- Controller** **The 'Brain':** Executes trajectories, computes kinematics & closes real-time feedback loops.
- Actuators** **The 'Muscles':** Convert electrical/fluid energy into physical movement (servos, steppers, pneumatics).

## 5. Robot Types & Applications

### Mechanical Architectures:

- **Manipulators:** Serial arms (Articulated, SCARA, Cartesian).
- **Mobile & Wheeled:** Differential drive, Omni, AGVs.
- **Legged:** Bipedal, quadrupeds, hexapods.
- **Autonomous Vehicles:** AUVs (Underwater), UAVs (Aerial drones).

### Primary Drivers (3 D's / U's):

- Dangerous** Nuclear cleanup, bomb disposal.
- Dull / Repetitive** High-speed assembly, spot welding.
- Dirty / Unwanted** Sewer inspection, toilet cleaning.

## 6. Robot Programming Methods

### ON-LINE METHODS (At the Robot)

- **Teach Pendant:** Handheld button box. Jog arm to waypoints & save; controller calculates PTP paths.
- **Lead-Through:** Manually guide arm by hand. Records continuous stream (60–80 pts/sec; high memory).

### OFF-LINE METHODS (Remote / Code)

- **Programming Languages:** Text code (AML, VAL, RobotStudio) without halting factory line.
- **Task-Level:** High-level goals ("pick item A") auto-compiled into trajectory commands.

## 7. Performance Metrics

- **Working Volume:** 3D envelope reachable by end-effector.
- **Speed & Accel:** Motion rate; trade-off with precision & payload.
- **Resolution:** Smallest commandable incremental step.
- **Accuracy:** Closeness between **commanded target** and **actual reached position** ( $|x_{\text{target}} - x_{\text{actual}}|$ ).
- **Repeatability:** Ability to return to the **exact same point** over repeated cycles ( $\sigma$ ).

### HIGH-YIELD EXAM DISTINCTION

- 💡 **Accuracy vs. Repeatability:** A robot can have **High Repeatability** (hits the identical spot 100 times in a row) but **Low Accuracy** (that spot is systematically 2 cm off target)!



## 1. Sensing & Transduction

- **Sensing:** Collecting information about the world.
- **Sensor:** Device mapping physical/chemical phenomena to quantitative signals.
- **Transduction Principle:** Converting one energy form into another.

| Sensor Type  | Transduction Mechanism         |
|--------------|--------------------------------|
| Thermistor   | Temp → Resistance ( $R$ )      |
| Photodiode   | Light → Current ( $I$ )        |
| Pyroelectric | Thermal rad. → Voltage ( $V$ ) |
| Humidity     | Moisture → Capacitance ( $C$ ) |
| LVDT         | Position → Inductance ( $L$ )  |
| Microphone   | Sound press. → Elect. signal   |

## 2. Sensor Classifications

### State Focus:

- **Proprioceptive** (Internal): Measures internal parameters (joint angle, wheel position, battery, tachometers/encoders/accelerometers).
- **Exteroceptive** (External): Environment & objects (proximity, vision, range).

### Energy Interaction:

- **Active:** Emits energy & reads reflection (Sonar, Radar, Modulated IR, LiDAR).
- **Passive:** Receives ambient energy only (Cameras, Pyroelectric, LDR).

### Contact & Modality:

- **Contact / Non-contact:** Touch (bumpers) vs. remote (optical, ultrasonic).
- **Visual / Non-visual:** Vision cameras vs. non-optical sensors.

## 3. Contact & Resistive Sensors

### A. FEELERS (TACTILE/CONTACT)

- **Whiskers:** Piano wire in metal hoop; deflection closes circuit; binary (0/1) output.
- **Bumpers & Guards:** Frame on microswitches; collision triggers switch; binary (0/1) output.

### B. RESISTIVE (VARIABLE R)

- **Bend Sensors:** Flexible strips where  $R$  increases as bent ( $10\text{ k}\Omega - 35\text{ k}\Omega$ ). *Uses:* Joint angles, wall-following, loads.
- **Potentiometers:** Linear/rotational variable resistors tracking sliding parts or rotating joint shafts.
- **Photocells (CdS / LDR):** Resistance shifts with light; highly non-linear, ideal for light tracking.

## 4. Capacitive vs. Inductive Sensors

### Capacitive (All Materials)

- Measures change in **dielectric**
- Detects **metals & non-metals** (liquids, plastics, powders)
- Senses through glass/plastic
- Sensitive to dust & humidity

#### APPLICATIONS

- **Level Detection:** Liquid/grains
- **Proximity:** Assembly lines
- **Touch Sensing:** Touchscreens

### Inductive (Metals Only)

- Emits AC magnetic field; senses **eddy currents**
- Detects **metals ONLY**
- Max on **ferrous** (iron/steel)
- Immune to oil, dust & water

#### APPLICATIONS

- **Position:** Metal part automation
- **Speed:** Rotating gears/shafts
- **Metal Detection:** Safety

## 5. Infrared (IR) Sensor Types

| Type                               | Principle & Trade-offs                                                                                                                                                              | Applications                                                                                                                   |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Intensity-Based Reflective</b>  | <ul style="list-style-type: none"> <li>• Measures reflected light</li> <li>• <b>Pro</b> Cheap, simple</li> <li>• <b>Con</b> Ambient &amp; color sensitive</li> </ul>                | <ul style="list-style-type: none"> <li>• Line-following</li> <li>• Wall-tracking</li> <li>• Break-beams</li> </ul>             |
| <b>Modulated IR Proximity</b>      | <ul style="list-style-type: none"> <li>• Flashes at <math>32 - 45\text{ kHz}</math></li> <li>• Receiver demodulates signal</li> <li>• <b>Pro</b> Immune to ambient light</li> </ul> | <ul style="list-style-type: none"> <li>• Proximity sensing</li> <li>• Remote controls</li> <li>• Obstacle avoidance</li> </ul> |
| <b>IR Triangulation PSD Ranger</b> | <ul style="list-style-type: none"> <li>• Emitter + lens + <b>PSD</b></li> <li>• Angle of return gives range</li> <li>• <b>Pro</b> Color &amp; light immune</li> </ul>               | <ul style="list-style-type: none"> <li>• Sharp GP2D02 (<math>10 - 80\text{ cm}</math>)</li> <li>• Distance map</li> </ul>      |

## 6. Ultrasonic Sensors & Noise

- **Principle:** Ultrasound burst ( $\approx 50\text{ kHz}$ ). Time-of-Flight ( $t$ ):

$$D = v \cdot t \implies d = \frac{v \cdot t}{2} \quad (v \approx 340\text{ m/s in air})$$

- **Limitations:**
  - **Bearing Uncertainty:** Wide beam spread (opening arc  $\approx 30^\circ$ ).
  - **Speed Latency:** Sound is slow (e.g.  $30\text{ m}$  round-trip takes  $\approx 200\text{ ms}$ ).
- **Noise Issues:** Acoustic noise & sensor crosstalk.

## 7. Laser Range Finders (LiDAR)

- **Performance:** Long range ( $2\text{ m} - 500\text{ m}$ ), high resolution ( $10\text{ mm}$ ), fast scans ( $13 - 40\text{ ms}$ ).
- **Robustness:** Narrow beam; immune to ambient light & specular reflections.

### HIGH-YIELD EXAM COMPARISON



- **Inductive vs. Capacitive:** Inductive senses **metals only**; Capacitive senses **all materials** (dielectric shift).
- **Ultrasonic vs. LiDAR:** Sonar has wide **bearing uncertainty** ( $\approx 30^\circ$ ) & latency; LiDAR provides millimeter angular precision.



## 1. Agency & Marr's 3-Level Theory

- **Agency:** Capacity of an entity (human or robot) to act intentionally, make choices, & exert control over actions and outcomes.
- **David Marr's Framework:** Analyzes information systems across 3 empirical levels:

| Level     | Question / Focus                     | Search & Rescue Robot Example                 |
|-----------|--------------------------------------|-----------------------------------------------|
| L1: Comp. | What goal / problem is solved?       | Locate trapped survivors in low-visibility    |
| L2: Algo. | What processes / IO represent it?    | Steer toward heat & CO <sub>2</sub> gradients |
| L3: Impl. | How physically realized in hardware? | Angle vectors to heat centroids; wheel motors |

**KEY EXAM CONCEPT** L1 & L2 are abstract & identical across biology/robots. Distinction appears **only at Level 3**.

## 2. Bio-Inspired Robotic Mappings

| Biological Model | Robotic Mapping         | Mechanism                                       |
|------------------|-------------------------|-------------------------------------------------|
| Bat Echolocation | Obstacle Avoidance      | Echo delay detection (ultrasonic)               |
| Ant Foraging     | Swarm Pathfinding       | Ant Colony Optim. (ACO), digital pheromones     |
| Bird Flocking    | Coordinated Drone Swarm | Reynolds Boids: alignment, cohesion, separation |
| Frog Tongue      | Fast Grasping Arm       | High-speed camera + snatch reflex               |
| Cuttlefish       | Adaptive Camouflage     | Pattern mapping onto E-skin                     |

## 3. Classifications of Behavior

**BEHAVIOR DEFINITION:** Direct mathematical mapping of **sensory inputs** to **motor actions** to achieve a task.

### 1. Reflexive (Stimulus-Response)

Hardwired / Fastest

- **Reflex** Lasts **only during stimulus** (knee-jerk, pupil reflex).
- **Taxis** **Directional steering** relative to stimulus:
  - *Tropotaxis:* Turtles to moonlight | - *Phototaxis:* Worms away from light | + *Rheotaxis:* Salmon upstream
- **FAP** Response **persists longer** than stimulus (spider web, nut caching, duckling imprinting).

### 2. Reactive Behaviors

Muscle Memory

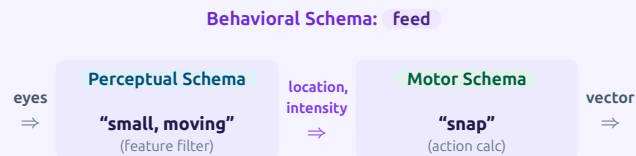
Learned skills consolidated to execute automatically without conscious thought (e.g. riding a bicycle).

### 3. Conscious Behaviors

Deliberative

Deliberate, goal-directed planning coordinating multiple sub-behaviors (e.g. puzzle solving, kit assembly).

## 4. Behavioral Schema Architecture



- **Frog Case Study:** Raw visual stimulus (eyes) is filtered into a clean percept (small, moving), passing coordinate ( $x, y$ ) and intensity to the motor schema to generate a tongue **snap vector**.

## 5. Schema Theory & OOP Robotics

- **Schema:** Foundational OOP template used for the **Reactive Layer** of autonomous robots.
- **Components of a Behavior Schema:**
  - **Perceptual Schema** Sensory feature extraction, threshold filtering & stimulus delays.
  - **Motor Schema** Computational procedure to produce physical action/movement vector.
- **Schema Instantiation (SI):** Creating a concrete runtime instance from a generic parameterized class template (e.g. applying specific handlebar height & seat position to a bike-riding schema).

## 6. Vector Action & S-R Notation

### Mathematical Formulations:

$$\{B : S \rightarrow R\} \quad \text{or} \quad B[S] = R$$

- $S$ : Perceptual function converting raw sensor feeds to **percept**.
- $R$ : Motor function converting percept into physical **action**.
- $B$ : Overarching behavioral system.

### Vector Field Combination:

- Directional stimuli (e.g. left & right eyes) combine via **Vector Summation** ( $\Sigma$ ):

$$V_{\text{resultant}} = v_{\text{left}} + v_{\text{right}}$$

### EXAM HIGHLIGHTS: LECTURE 03

- **Reflex vs. FAP Duration:** Reflex stops as soon as stimulus stops; FAP continues for longer duration.
- **Taxis Examples:** Positive Rheotaxis (salmon upstream), Tropotaxis (sea turtles to moonlight), Negative Phototaxis (earthworms).
- **Marr's Level 3:** L1 & L2 are identical in biology & robotics; differences emerge **only at Level 3**.

## 1. LEGO Mindstorms EV3 Architecture

- **Processing Unit:** ARM9 @ 300 MHz, 16 MB Flash, 64 MB RAM.
- **Operating System:** Embedded Linux-based OS.
- **Sensor Ports (1-4):** 4 Input ports supporting Analog & high-speed Digital UART up to 460.8 kbit/s.
- **Motor Ports (A-D):** 4 Output ports with built-in optical rotation encoders for position/speed feedback.
- **I/O & Expansion:**
  - High-speed USB (480 Mbit/s) + Host daisy-chaining (up to 3 levels), WiFi dongle, & USB storage.
  - Micro SD-card slot (supports up to 32 GB).
- **Display & UI:** 178 × 128 monochrome LCD matrix, 6 backlit navigation/debugging buttons.
- **Wireless Comms:** Bluetooth v2.1 + EDR, Apple/Android smart device protocol.

## 2. Platform Generations & Modules

| Platform   | CPU / OS   | Key Ports & Features                    |
|------------|------------|-----------------------------------------|
| RCX (1998) | Hitachi H8 | 3 In / 3 Out; Optical IR tower comms    |
| NXT (2006) | ARM7 + AVR | 4 In / 3 Out; Bluetooth, 100 × 64 LCD   |
| EV3 (2013) | ARM9 Linux | 4 In / 4 Out; Daisy-chain, USB Host, SD |

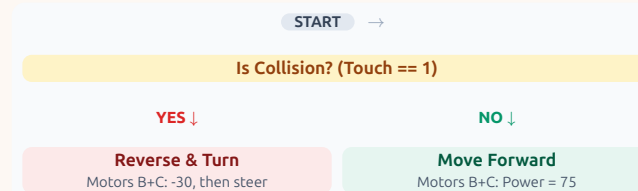
### Sensor & Actuator Modules:

- **Motors:** Large (high torque) & Medium (high RPM) with integrated tachometers.
- **Sensors:** Touch (binary limit), Color (ambient/reflected/RGB), IR Seeker/Beacon, Ultrasonic.

## 3. Step 1: Avoid Collision (Reflex)

### PURE REFLEXIVE BEHAVIOR

- **Objective:** Move continuously; instantly back up and pivot when a physical impact occurs.

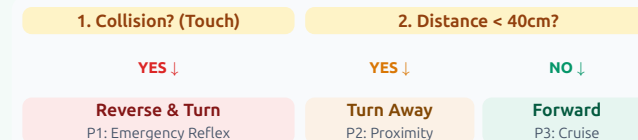


- **Collision == True:** Reverse (−30, 2 rot) → steer turn (−45/30, 2 rot).
- Direct hardware-to-motor S-R loop without state memory.

## 4. Step 2: Collision + Obstacle Avoidance

### HIERARCHICAL PRIORITY ARBITRATION

- Combines contact (Touch) and non-contact (Proximity/Force) into a 2-tier reactive hierarchy:

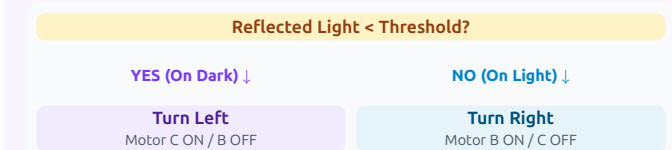


- **P1 (Touch):** Emergency physical recovery (overrides all).
- **P2 (Proximity):** Smooth steering turn away before contact.
- **P3 (Default):** Drive straight ahead.

## 5. Step 3a: 1-Sensor Line Following

### BANG-BANG / EDGE TRACKING

- Tracks the **boundary edge** between light and dark surfaces using 1 photo/color sensor.



- Ref < Th (On black) → Steer outside (Turn Left).
- Ref ≥ Th (On white) → Steer inside (Turn Right).
- **Control Characteristic:** 2-state discrete bang-bang controller; causes high-frequency zigzag oscillation along line edge.

## <> 6. Reactive Control Summary

| Stage   | Sensor Used     | Behavior Pattern           |
|---------|-----------------|----------------------------|
| Step 1  | Touch Switch    | Emergency reflex backoff   |
| Step 2  | Touch + IR/Dist | Layered priority avoidance |
| Step 3a | Color/Light     | 1-Sensor edge tracking     |

### EXAM HIGHLIGHTS: LECTURE 04

- **EV3 vs NXT/RCX:** EV3 introduced Linux OS, ARM9 @ 300MHz, daisy-chaining, USB host, & 4th motor port.
- **1-Sensor Line Following:** Always tracks an **edge** (boundary transition), never the centerline itself.
- **Priority Arbitration:** Safety/impact reflexes always take precedence over non-contact navigation.

## 1. Control Paradigm Taxonomy

- Core Philosophy:** How sensing, decision-making, and execution interact.

| Paradigm                    | Information Flow & Focus                                       |
|-----------------------------|----------------------------------------------------------------|
| Hierarchical / Deliberative | Sense → Plan → Act<br>Global world model, top-down plan        |
| Reactive / Behavior-Based   | Sense → Act<br>Fast direct S-R loops, no world model           |
| Hybrid                      | Plan → [Sense → Act]<br>Deliberative mission + reactive reflex |

### Hierarchical (Sense-Plan-Act):



- 'Eyes Closed' Execution:** Robot senses once, computes entire plan, then executes directives without re-evaluating until loop restarts.

## 2. Monolithic Sensing & Architecture

- Monolithic Sensing:** All sensor observations fuse into **one global data structure** (World Model) accessed by planner:
  - A Priori** Pre-loaded maps/building layout.
  - Sensing Info** Current state & localization ("in NW hallway").
  - Cognitive** Mission goals (deliver item to Room 118).

### Autonomous Vehicle Pipeline:

| Stage      | Sub-Modules & Tasks                          |
|------------|----------------------------------------------|
| Sensors    | Cameras, Radar, LiDAR, GPS / IMU             |
| Perception | Obstacle Detection, Free Space, Localisation |
| Planning   | Route → Behavioral → Trajectory              |
| Control    | PID, MPC → Drive-By-Wire (DBW) Actuators     |

## 3. Shakey & STRIPS Planning (6-Step Algorithm)

- Shakey (1967-69, SRI / DARPA):** First AI mobile robot using **Means-Ends Analysis** (reduces state difference  $\Delta$ ).

### The 6 Steps in Executing STRIPS

### Recursive Stack

- Compute difference:**  $\Delta = \text{Goal} - \text{Initial State}$ . If  $\Delta = \emptyset$ , terminate.
- Select Operator:** If  $\Delta \neq \emptyset$ , pick 1st operator in Difference Table whose **add-list** negates  $\Delta$ .
- Examine Preconditions:** Check if variable bindings evaluate all preconditions to **TRUE**.
- Recurse on Subgoal:** If FALSE, push original goal to stack, set 1st FALSE condition as **new subgoal**, & recurse.
- Push to Plan Stack:** When all preconditions match, push operator to plan stack & update world model copy.
- Pop & Resume:** Return to parent operator with failed condition to apply it or recurse on next condition.

## 4. STRIPS Predicates & Operator Anatomy

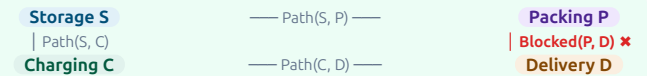
- Predicates:** UPPERCASE statements evaluating to  $\frac{\text{TRUE}}{\text{FALSE}}$  (e.g.  $\text{INROOM}(\text{IT}, R_1)$ ,  $\text{CONNECTS}(D_1, R_1, R_2)$ ).

| Operator                 | Preconditions                                                                                           | Add-List                        | Delete-List                     |
|--------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------|---------------------------------|
| <b>GOTODOOR (IT, dx)</b> | $\text{INROOM}(\text{IT}, r_k)$<br>$\text{CONNECT}(d_x, r_k, r_m)$                                      | $\text{NEXTTO}(\text{IT}, d_x)$ | —                               |
| <b>GOTHRU (IT, dx)</b>   | $\text{CONNECT}(d_x, r_k, r_m)$<br>$\text{NEXTTO}(\text{IT}, d_x)$<br>$\text{STATUS}(d_x, \text{OPEN})$ | $\text{INROOM}(\text{IT}, r_m)$ | $\text{INROOM}(\text{IT}, r_k)$ |

- Generated Plan:** Stack sequence:  $\text{GOTODOOR}(\text{IT}, D_1) \rightarrow \text{GOTHRUDOOR}(\text{IT}, D_1)$ .

## 5. Warehouse STRIPS Case Study

**Warehouse Layout:** Locations Charging (C), Storage (S), Packing (P), Delivery (D).



| Operator             | Preconditions                                                                 | Add-List                             | Delete-List                          |
|----------------------|-------------------------------------------------------------------------------|--------------------------------------|--------------------------------------|
| <b>Move(x, y)</b>    | $\text{Path}(x, y) \neg \text{Blocked}(x, y)$<br>$\text{At}(\text{Robot}, x)$ | $\text{At}(\text{Robot}, y)$         | $\text{At}(\text{Robot}, x)$         |
| <b>PickUp(p, x)</b>  | $\text{At}(p, x)$ $\text{At}(\text{Robot}, x)$<br>$\text{HandEmpty}$          | $\text{Holding}(p)$                  | $\text{At}(p, x)$ $\text{HandEmpty}$ |
| <b>PutDown(p, x)</b> | $\text{Holding}(p)$ $\text{At}(\text{Robot}, x)$                              | $\text{At}(p, x)$ $\text{HandEmpty}$ | $\text{Holding}(p)$                  |

- Obstacle Contingency:** Since  $\text{Blocked}(P, D)$  is TRUE, planner avoids path  $S \rightarrow P \rightarrow D$  and selects detour:

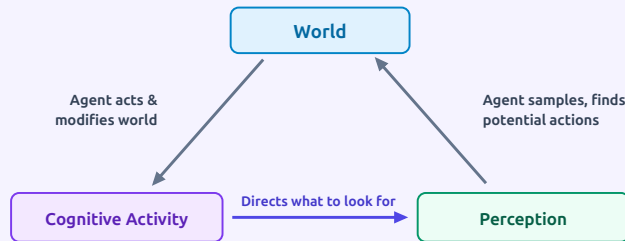
**Detour Plan:**  $\text{Move}(C, S) \rightarrow \text{PickUp}(P_1, S) \rightarrow \text{Move}(S, C) \rightarrow \text{Move}(C, D) \rightarrow \text{PutDown}(P_1, D)$

### HIGH-YIELD EXAM DISTINCTION (LECTURE 05)

- Deliberative Pros vs Cons:** Pro: Computes optimal, goal-directed solutions. Con: Fragile under uncertainty, high computational complexity, slow to react to dynamic changes. **STRIPS Add/Delete Rule:** World state updates **only** by deleting predicates in Delete-List and appending predicates in Add-List upon successful precondition binding.

## 1. Action-Perception & Ecological Approach

- **Action-Perception Cycle:** Continuous closed feedback loop where perception guides actions, and actions alter the environment.



- **Gibson's Ecological Approach:**
  - **Core Tenet** "... the world is its own best representation."
  - Perception evolved **only to support survival actions** in an ecological niche, not to build abstract 3D maps.

## 2. Affordances & Perceptual Systems

- **Affordance:** Directly perceivable potentiality for action provided by the environment (e.g. a chair affords **sitability**).
- **Direct Perception:** Requires **no memory, inference, or interpretation** → near-instantaneous execution.
  - **Visual: Optic Flow** Pattern of apparent visual motion; expanding outward from center gives **time-to-contact** (diving gannets, landing).
  - **Non-Visual** Sound pitch change when filling a container indicates fullness without knowing cavity dimensions.

### Direct Perception (Gibsonian)

- Primitive brain structures - Local world models - Fast, lightweight affordances

### Recognition (Cognitive)

- High cognitive brain - Global world models - Top-down internal templates ("my cup vs your cup")

## 3. Behavior Acquisition Spectrum

| Type               | Biological Model   | Mechanism & Strategy                                                                            |
|--------------------|--------------------|-------------------------------------------------------------------------------------------------|
| Innate             | Arctic Tern chicks | Born with rule: "Peck at largest red blob" → triggers parent feeding reflex.                    |
| Sequence of Innate | Digger Wasp        | Mating → internal state triggers nest building → visual nest triggers egg laying.               |
| Innate with Memory | Baby Bees          | Innate flight exploration in expanding arcs to memorize hive appearance and approach vectors.   |
| Learned            | Lion Cubs          | Born with zero hunting behaviors; complex multi-stage skills (stalk, chase) learned over years. |

## 4. Innate Releasing Mechanisms (IRM)

- **IRM Definition:** Theoretical construct where specific sensory triggers (**sign stimuli / releasers**) activate fixed behaviors.
- **Releaser as Control Signal:** Acts as a Boolean latch turning a behavior ON or OFF. If not released, behavior produces zero output.
- **Compound Releasers:** Logical combination of **external stimuli** and **internal motivation** (e.g. Hungry  $\wedge$  FoodPresent).

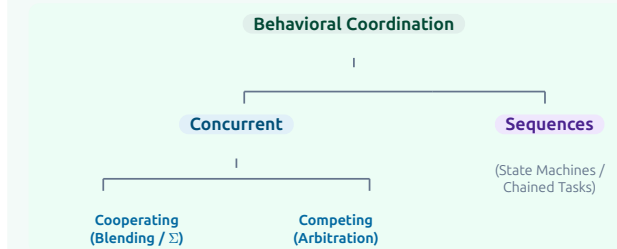
### Execution Modes & State Control

- **Implicit Chaining:** Execution order emerges dynamically from which releasers are TRUE.
- **Inhibition & FAP Persistence:** Prevents rapid flickering between states (e.g. flee vs feed) by locking behavior for time  $T$ .
- **Explicit Sequences:** State-machine order where each behavior runs to completion before advancing.

### Coordination

## 5. Coordination & Concurrent Behaviors

### Behavioral Coordination Architecture:



### 3 Types of Concurrent Interactions:

- **1. Equilibrium:** Behaviors balance each other out (e.g. **Squirrel** hesitant between food lure and fear of human).
- **2. Dominance (Winner-Take-All):** One behavior completely suppresses others (e.g. sleep vs feed, flight over forage).
- **3. Cancellation:** Opposing stimuli cancel each other, leaving displacement behavior (e.g. **Male Stickleback** fish building a new nest when territory defense & attack cancel).

### HIGH-YIELD EXAM DISTINCTION (LECTURE 06)

- **Design Hierarchy:** Always solve behavior using **Affordances** (direct perception) first if possible; resort to **Recognition** (world models) only when cognitive identification is strictly required. **IRM Control vs Perceptual Filter:** An IRM's releaser decides **WHEN** to run; perceptual schema filters **WHAT** to track.